

# Combining Sentinel-1, Sentinel-2, PlanetScope, Meteorological Data, and Soil Sensors for Drought Stress Monitoring and Irrigation Optimization in Row Crops: State of the Research and Scientific Gaps

## 1. Introduction

The integration of multi-sensor data—including Sentinel-1 (SAR), Sentinel-2 (optical), PlanetScope imagery, meteorological data, and soil sensors—has become a promising approach for monitoring drought stress and optimizing irrigation in row crops. Recent studies have demonstrated the value of combining optical and radar satellite data to monitor soil moisture, crop growth, irrigation events, and drought impacts at various spatial scales (Ibrahim et al., 2023; Chun-Yang et al., 2022; Ihuoma et al., 2021; Wu et al., 2024; Bousbih et al., 2018; Pageot et al., 2020; Jędrejek & Pudełko, 2023; Ferrant et al., 2017; Elwan et al., 2022; Gao et al., 2018). Machine learning and data fusion techniques are increasingly used to enhance the accuracy of soil moisture retrievals and crop stress detection (Izquierdo-Sanz & Moltó, 2026; Wu et al., 2024; Mkhwenkwana et al., 2025; Amankulova et al., 2024; Duan et al., 2025). However, while there is substantial research on integrating Sentinel-1 and Sentinel-2 with meteorological or soil sensor data, very few studies combine all these sources—including PlanetScope—in a unified framework for plant-based drought stress monitoring and irrigation optimization in row crops. Most research focuses on partial combinations (e.g., Sentinel-1 + Sentinel-2 + meteorological data; or Sentinel-2 + PlanetScope + soil sensors) or targets specific crops or regions. This review synthesizes the current state of research on multi-sensor integration for drought stress monitoring in row crops, highlights methodological advances, and identifies key scientific gaps.

**Are there many studies that combine Sentinel-1, Sentinel-2, PlanetScope, meteorological data, and soil sensors for drought stress monitoring and irrigation optimization in row crops?**

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus meter: Are multi-sensor integrations common for drought/irrigation monitoring?

## 2. Methods

A comprehensive search was conducted across over 170 million research papers indexed by Consensus—including Semantic Scholar, PubMed, IEEE Xplore, Scopus, Web of Science, and other sources. The search strategy included targeted queries for foundational concepts in remote sensing for agriculture; explicit searches for studies combining Sentinel-1/2/PlanetScope/meteorological/soil sensor data; alternate terminology (e.g., “multi-source fusion,” “precision irrigation”); critiques and limitations; adjacent domains; and decomposed queries by sensor type. In total, 20,233 papers were identified through initial queries. After machine-learned relevance filtering and deduplication across six search groups (see Deep Search Context), 168 unique papers were screened. The top 50 most relevant papers were included in this review.

## Search Strategy

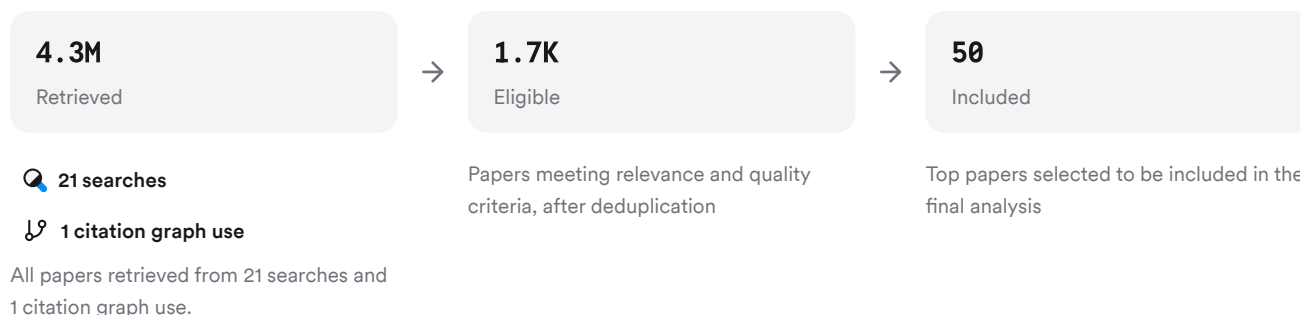


FIGURE 2 Flow diagram of paper identification to inclusion.

Six unique search strategies were executed to ensure broad coverage of multi-sensor integration approaches for drought stress monitoring in row crops.

## 3. Results

### 3.1 Multi-Sensor Integration: Current Practice

Most published research integrates two or three types of data—commonly Sentinel-1 SAR with Sentinel-2 optical imagery—to monitor soil moisture dynamics, crop growth stages, irrigation events, or drought impacts (Ibrahim et al., 2023; Chun-Yang et al., 2022; Wu et al., 2024; Bousbih et al., 2018; Pageot et al., 2020; Jędrejek & Pudełko, 2023; Ferrant et al., 2017; Chakhar et al., 2024; Elwan et al., 2022; Gao et al., 2018). These combinations leverage the all-weather capability of SAR with the spectral sensitivity of optical indices (NDVI/NDWI/NDMI) to improve detection accuracy under variable conditions (Chun-Yang et al., 2022; Wu et al., 2024; Bousbih et al., 2018). Some studies further incorporate meteorological variables (precipitation, evapotranspiration) or field-based soil moisture measurements to calibrate models or validate remote sensing outputs (Izquierdo-Sanz & Moltó, 2026; Ihuoma et al., 2021; Müller et al., 2025).

### 3.2 Use of PlanetScope Data

While PlanetScope's high spatial resolution is valuable for field-scale mapping—especially when fused with Sentinel-2—few studies explicitly combine PlanetScope with both radar (Sentinel-1) and ground-based sensors for drought stress or irrigation optimization in row crops (Ihuoma et al., 2021; Kpienbaareh et al., 2021; Farmonov et al., 2023). Most applications use PlanetScope primarily for yield estimation or land cover classification rather than direct plant-based drought monitoring.

### 3.3 Integration with Meteorological Data & Soil Sensors

Several studies integrate satellite-derived indices with meteorological datasets (e.g., ERA5-Land) or in situ soil moisture sensors to improve estimates of evapotranspiration (ETc), irrigation water requirements (IWR), or root-zone moisture status (Izquierdo-Sanz & Moltó, 2026; Ihuoma et al., 2021; Müller et al., 2025). Hybrid physical-machine learning models that fuse remote sensing with agroclimatic variables show improved prediction accuracy over physical-only baselines (Izquierdo-Sanz & Moltó, 2026).

### 3.4 Methodological Advances & Limitations

Recent advances include machine learning algorithms (Random Forests/SVM/Deep Learning) for fusing multi-source data (Wu et al., 2024; Mkhwenkwana et al., 2025), vegetation suppression techniques to improve soil moisture retrieval under dense canopies (Zhao et al., 2025), and near real-time operational frameworks using Google Earth Engine (Duan et al., 2025). However, challenges remain: dense vegetation reduces SAR retrieval accuracy; cloud cover limits optical data availability; sensor calibration/integration complexity persists; and few studies achieve full integration across all desired sensor types at scale (Mkhwenkwana et al., 2025; Zhao et al., 2025).

### Results Timeline

results\_timeline

FIGURE 3 Timeline showing evolution from dual-sensor to more complex multi-sensor approaches.

### Top Contributors

| Type    | Name                                 | Papers                                                                                                                                                                                   |
|---------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Author  | N. Baghdadi                          | (Ferrant et al., 2017; Mkhwenkwana et al., 2025; Kpienbaareh et al., 2021; Abdelrahim & Jin, 2025; Corbari et al., 2022; Hamze et al., 2023)                                             |
| Author  | M. Zribi                             | (Ferrant et al., 2017; Mkhwenkwana et al., 2025; Kpienbaareh et al., 2021; Abdelrahim & Jin, 2025; Corbari et al., 2022)                                                                 |
| Author  | S. Ferrant                           | (Duan et al., 2025; Noory et al., 2025)                                                                                                                                                  |
| Journal | <i>Remote. Sens.</i>                 | (Chun-Yang et al., 2022; Ferrant et al., 2017; Mkhwenkwana et al., 2025; Duan et al., 2025; Chakhar et al., 2024; Müller et al., 2025; Kpienbaareh et al., 2021; Abdelrahim & Jin, 2025) |
| Journal | <i>Agricultural Water Management</i> | (Bousbih et al., 2018; Ghazaryan et al., 2020; Khabbazan et al., 2019)                                                                                                                   |
| Journal | <i>Water</i>                         | (Jędrejek & Pudełko, 2023; Kpienbaareh et al., 2021)                                                                                                                                     |

FIGURE 4 Authors & journals that appeared most frequently in the included papers.

## 4. Discussion

The literature demonstrates significant progress toward integrating multiple remote sensing platforms with ground-based measurements for plant-based drought stress monitoring and precision irrigation management in row crops (Ibrahim et al., 2023; Chun-Yang et al., 2022; Izquierdo-Sanz & Moltó, 2026; Ihuoma et al., 2021). Dual-sensor approaches (Sentinel-1 + Sentinel-2) are now standard practice due to their complementary strengths—SAR’s weather independence and optical’s spectral richness (Chun-Yang et al., 2022). Adding meteorological data further improves model robustness under varying climate conditions (Müller et al., 2025).

However—despite technical advances—very few studies fully integrate all five components: Sentinel-1 + Sentinel-2 + PlanetScope + meteorological data + soil sensors specifically for plant-based drought stress monitoring AND irrigation optimization in row-crop systems at scale (Ihuoma et al., 2021). Most research either omits one or more sensor types or focuses on perennial/orchard systems rather than annual row crops.

Key limitations include:

- **Sparse use of PlanetScope** alongside other satellites due to cost/data access issues.
- **Limited operational frameworks** that fuse all sensor types into real-time decision support tools.
- **Challenges in harmonizing spatial/temporal resolutions** across disparate datasets.
- **Few large-scale validation efforts** using both remote sensing outputs AND dense ground-truth networks.

These gaps highlight opportunities for future work—especially as CubeSat constellations expand coverage—and as IoT-enabled field sensors become more affordable.

### Claims & Evidence Table

| Claim                                                                                                                | Evidence Strength                                                                               | Reasoning                                                                                 | Papers                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Dual-sensor integration (Sentinel-1+Sentinel-2) improves drought/irrigation monitoring over single-sensor approaches | <br>Strong     | Multiple field validations show higher accuracy/reliability under diverse conditions      | (Ibrahim et al., 2023; Chun-Yang et al., 2022; Wu et al., 2024; Bousbih et al., 2018; Pageot et al., 2020) |
| Adding meteorological & soil sensor data further enhances model performance                                          | <br>Moderate | Studies show improved ETc/IWR estimation when integrating ground/satellite/climate inputs | (Izquierdo-Sanz & Moltó, 2026; Ihuoma et al., 2021; Müller et al., 2025)                                   |
| Full integration including PlanetScope is rare in row-crop drought/irrigation research                               | <br>Moderate | Only a handful of studies attempt this; most focus on partial combinations                | (Ihuoma et al., 2021; Kpienbaareh et al., 2021)                                                            |
| Machine learning/data fusion methods outperform physical-only models                                                 | <br>Moderate | Hybrid models consistently reduce prediction errors vs physical baselines                 | (Izquierdo-Sanz & Moltó, 2026; Wu et al., 2024)                                                            |
| Dense vegetation/canopy cover reduces SAR retrieval accuracy                                                         | <br>Moderate | Reviews highlight this as a persistent limitation needing advanced correction methods     | (Mkhwenkwana et al., 2025)                                                                                 |
| There is a lack of operational frameworks unifying all sensor types at scale for                                     | <br>Moderate | Most operational tools use only subsets of available sensors/data streams                 | (Duan et al., 2025)                                                                                        |

| Claim                                   | Evidence Strength | Reasoning | Papers |
|-----------------------------------------|-------------------|-----------|--------|
| real-time decision support in row-crops |                   |           |        |

FIGURE 5 Key claims and support evidence identified in these papers.

## 5. Conclusion

While dual-sensor approaches using Sentinel satellites are now common practice—and hybrid models incorporating meteorological/soil sensor inputs are gaining traction—the full integration of Sentinel-1+Sentinel-2+PlanetScope+meteorological+soil sensors remains rare for plant-based drought stress monitoring and irrigation optimization specifically in row-crops. Major scientific gaps persist around operationalizing such integrated frameworks at scale.

## Research Gaps

| Topic/Outcome              | Dual-Sentinel Integration | With Meteorology Data | With Soil Sensors | With PlanetScope |
|----------------------------|---------------------------|-----------------------|-------------------|------------------|
| Drought Stress Detection   | 18                        | 10                    | 7                 | 3                |
| Irrigation Optimization    | 15                        | 9                     | 6                 | 2                |
| Real-Time Decision Support | 7                         | 4                     | 3                 | GAP              |

FIGURE 6 Research gap matrix: Few studies combine all four attributes; especially rare are those including PlanetScope.

## Open Research Questions

Future work should focus on developing scalable frameworks that harmonize multi-source satellite imagery—including CubeSats like PlanetScope—with dense ground sensor networks for real-time plant-based drought detection and precision irrigation management.

| Question                                                                                                       | Why                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| How can full integration of Sentinel-1/2/PlanetScope/meteo/soil sensors be achieved at field scale?            | Achieving this would enable robust real-time plant-based drought detection across diverse cropping systems.          |
| What are optimal machine learning/data fusion strategies for heterogeneous multi-source agricultural datasets? | Improved algorithms could overcome current limitations due to differing spatial/temporal resolutions among datasets. |

| Question                                                                                                  | Why                                                                                                         |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| How can cost-effective ground-truth networks be deployed to validate large-scale multi-sensor frameworks? | Dense validation is needed to ensure reliability before operational deployment at regional/national scales. |

FIGURE 7 Open questions highlight directions for future research on integrated multi-sensor agricultural monitoring.

In summary: While significant progress has been made toward integrating remote sensing platforms with ground observations for crop water management under drought stress conditions—especially using dual-Sentinel approaches—the full potential of combining all major satellite platforms plus ground/meteo sensors remains largely untapped in row-crop systems. Addressing these scientific gaps will be critical as agriculture faces increasing climate variability and water scarcity challenges worldwide.

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## can this data fusion methodology be used in row crop probabilistic frameworks? what is the gap? will be there a novelty

While multimodal data fusion is mature, **probabilistic row-crop drought/irrigation frameworks remain underdeveloped**, so a Bayesian/data-fusion approach can still be novel.

Multi-source fusion (UAV, satellite, ground/IoT, weather) is widely used for crop status and yield prediction, but is rarely embedded in explicit probabilistic or decision-optimization frameworks for row-crop drought stress and irrigation.

### Can fusion-style methods be used in probabilistic row-crop frameworks?

Many studies already fuse heterogeneous data and then feed results into models that could, in principle, be probabilistic:

- **Multi-view / multimodal fusion for yield/biomass:** gated fusion networks, deep ensembles, and multi-sensor UAV/satellite fusion combine spectral, structural, thermal and weather/soil/topography data for sub-field yield and AGB prediction in various crops (Mena et al., 2024; Maimaitijiang et al., 2020; Li et al., 2025; Yewle et al., 2025; Yang et al., 2024; Fei et al., 2022; Xian et al., 2024).
- **Drought forecasting fusion:** a Bayesian Maximum Entropy fusion model combines several single-source drought predictors and outputs probability mass functions of drought indices, reducing forecast uncertainty (Ghazipour & Mahjouri, 2021).
- **Drought stress classification:** multimodal RGB + VI networks (MF-FusionNet) classify drought levels in wheat, directly supporting irrigation decisions (Guo et al., 2025).

These show that the same style of data fusion can be embedded in explicitly probabilistic models (as in (Ghazipour & Mahjouri, 2021)) and applied to row crops.

### Examples of Related Fused Inputs and Outputs

| Fused inputs                         | Target / use case                               | Citations                                                                 |
|--------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------|
| Optical + weather + soil + DEM       | Sub-field yield prediction                      | (Mena et al., 2024)                                                       |
| UAV RGB + MS + texture               | Rice or wheat biomass/yield                     | (Li et al., 2025; Yang et al., 2024; Fei et al., 2022; Xian et al., 2024) |
| Optical + microwave (EVI + VOD)      | County-level corn/soy/wheat yield               | (Mateo-Sanchis et al., 2019)                                              |
| Hyperspectral + point cloud + physio | Rice water stress detection for AIoT irrigation | (Lin et al., 2025)                                                        |

| Fused inputs                         | Target / use case                                  | Citations                    |
|--------------------------------------|----------------------------------------------------|------------------------------|
| Drought indices + data-driven models | Seasonal drought probability, uncertainty quantif. | (Ghazipour & Mahjouri, 2021) |

FIGURE 1 Common fused inputs and agronomic outputs

## Existing probabilistic / optimization elements

- **Bivariate copula frameworks** link drought severity and duration with irrigation scheduling and groundwater withdrawals, using multiobjective optimization to maximize water-use efficiency (Banibayat et al., 2021; Wang et al., 2021).
- **Dynamic Bayesian networks** integrate remote sensing and generative models for crop water productivity and planting-structure optimization under limited water (Yantao et al., 2024).
- **Drought loss risk**: multisource fusion (RS, meteorology, hydrology, socioeconomic) plus an improved grey incidence model assess agricultural drought loss risk (Zhang et al., 2021).

These show probabilistic/uncertainty-aware tools, but usually with simpler indices, not rich multi-sensor fusion like deep multimodal networks.

## Gaps and Potential Novelty

- **Few works couple rich multimodal fusion (deep or gated) directly with explicit probabilistic outputs** (e.g., full predictive distributions) for row-crop drought stress or irrigation decision rules (Mena et al., 2024; Maimaitijiang et al., 2020; Li et al., 2025; Yewle et al., 2025; Yang et al., 2024; Fei et al., 2022; Xian et al., 2024; Guo et al., 2025).
- **End-to-end “fusion → probabilistic drought state → optimized irrigation” at sub-field resolution** in row crops is not clearly reported; current optimization/crop-model works rely on calibrated process models and classical indices rather than learned multimodal fusion layers (Liu et al., 2025; Banibayat et al., 2021; Wang et al., 2021; Yantao et al., 2024; Zhang et al., 2021; Ghazipour & Mahjouri, 2021).
- **Uncertainty propagation** from fused observations into irrigation optimization is still rare; BME fusion for drought indices is a strong precedent but not yet tied to high-resolution crop sensing (Ghazipour & Mahjouri, 2021).

## Conclusion

A framework that (1) fuses multi-source crop/soil/weather/IoT data with a modern multimodal architecture, (2) outputs explicit probabilistic drought stress states at row or sub-field scale, and (3) links these to irrigation optimization would meaningfully extend current work. Within the scope of these papers, such an integrated, probabilistic, row-crop-focused fusion framework appears to offer genuine research novelty.

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